E-COMMERCE SALES & PROFIT ANALYSIS

A Business Intelligence & Sales Analytics Project

Using Power BI

PROJECT DETAILS	DETAILS
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Academic Year	2026–2027

CERTIFICATE

This is to certify that Bhomesh Doyal, student of BVOC AIDS Batch A, Thakur College of Engineering & Technology, has successfully completed the project titled “E-Commerce Sales & Profit Analysis: A Business Intelligence & Sales Analytics Project Using Power BI” under my guidance during the academic year 2026–2027.

The work presented in this report is submitted as part of the academic requirements of the program and demonstrates the application of data preparation, data modelling, DAX calculations, visualization and business analysis using Microsoft Power BI.

Place: _____ Date: _____

Signature: _____

DECLARATION

I hereby declare that the project report entitled “E-Commerce Sales & Profit Analysis: A Business Intelligence & Sales Analytics Project Using Power BI” is my original academic work prepared for the purpose of demonstrating practical skills in Business Intelligence, data analysis and dashboard development.

The data preparation, calculations, dashboard design and interpretation presented in this report have been prepared for this project. Where external datasets, software documentation or learning resources have been used, they have been acknowledged in the references section.

The project includes sales and profit analysis because the supplied e-commerce dataset contains both Total_Sales and Profit fields. The conclusions presented are limited to the information available in the dataset.

Student Signature: _____

Name: Bhomesh Doyal

Date: _____

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Thakur College of Engineering & Technology for providing the academic environment and resources required to complete this project.

I am thankful to my project guide and faculty members for their guidance, suggestions and support throughout the development of the Business Intelligence dashboard.

I also acknowledge the datasets, software documentation and learning resources that supported the data preparation, Power BI modelling, DAX calculations and visualization work carried out in this project.

ABSTRACT

This project presents a Business Intelligence solution for analysing e-commerce sales and profitability using Microsoft Power BI. The objective is to transform transactional sales data into an interactive dashboard that helps users understand sales performance, profit, order volume, product-category performance, regional contribution and customer-segment behaviour.

The dataset contains transaction-level information including country, customer name, customer segment, discount percentage, order date, order ID, payment method, product category, product name, profit, quantity, region, shipping cost, total sales and unit price. The project applies data validation, calendar modelling, relationships, DAX measures and interactive visualizations.

A dedicated Calendar table is connected to the Ecommerce_Sales table through Order_Date. Core measures include Total Sales, Total Profit, Total Orders, Total Quantity, Profit Margin and Average Order Value. The dashboard uses KPI cards, a monthly sales trend, regional sales and profit charts, product-category analysis, a Top 10 Products visual, slicers and a donut chart for profit contribution by product category.

The resulting Power BI report provides an executive view of e-commerce performance and supports business decisions related to product categories, regions, customer segments, pricing, profitability and sales trends.

Keywords: Business Intelligence, Power BI, DAX, E-Commerce Analytics, Sales Analytics, Profit Analysis, Dashboard, Data Visualization, KPI, Regional Analysis

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CHAPTER 1 — INTRODUCTION

1.1 Background

E-commerce businesses generate large volumes of transactional data through online orders. However, raw transaction records are difficult to interpret without structured analysis. Business Intelligence converts these records into meaningful measures, trends and visual patterns that support faster and more consistent decision-making.

For an e-commerce business, management needs to understand total sales and profit, product performance, regional contribution, customer-segment behaviour and changes in sales over time. These questions form the analytical foundation of this project.

1.2 Problem Statement

The raw e-commerce dataset contains detailed transaction records but does not directly provide an executive view of business performance. Without an analytical model, it is difficult to identify revenue drivers, profitable regions, product-category opportunities and sales trends. The project therefore develops a Power BI dashboard that converts transaction-level sales data into an interactive management reporting solution.

1.3 Project Aim

To design and implement an interactive Power BI Business Intelligence dashboard that analyses e-commerce sales and profit performance and converts transactional data into actionable business insights.

1.4 Objectives

- Prepare and validate the e-commerce sales dataset for analysis.
- Build a reusable Calendar table for time-based reporting.
- Create a relational data model connecting Calendar and Ecommerce_Sales.
- Develop DAX measures for sales, profit, orders, quantity, profit margin and average order value.
- Identify high-performing and low-performing product categories and products.
- Analyse regional sales and profit performance.
- Compare customer-segment and product-category performance.
- Visualize monthly sales trends and important business patterns.
- Present findings through a professional interactive Power BI dashboard.
- Translate analytical results into practical business recommendations.

1.5 Scope

The project focuses on sales and profit performance represented in the supplied e-commerce transaction data. It covers sales, profit, orders, quantity, products, product categories, regions, customer segments, discounts, shipping cost and dates. Advanced customer lifetime value, inventory forecasting and operational cost analysis are outside the scope unless the required data becomes available.

CHAPTER 2 — DATASET & BUSINESS REQUIREMENTS

2.1 Dataset Overview

The project uses an e-commerce sales dataset imported into Microsoft Power BI. The principal table used for analysis is named Ecommerce_Sales. Each record represents a transaction or sales-line entry containing customer, product, order and financial information.

2.2 Dataset Fields

Field	Type / Role	Business Meaning
Country	Text / Category	Geographical country
Customer_Name	Text	Customer name
Customer_Segment	Text / Category	Customer classification
Discount_Percent	Decimal Number	Discount applied
Order_Date	Date	Order date
Order_ID	Text / Identifier	Unique order identifier
Payment_Method	Text / Category	Payment method
Product_Category	Text / Category	Product category
Product_Name	Text / Category	Product name
Profit	Decimal Number	Profit generated
Quantity	Whole Number	Units sold
Region	Text / Category	Sales region
Shipping_Cost	Decimal Number	Shipping cost
Total_Sales	Decimal Number	Sales value
Unit_Price	Decimal Number	Price per unit

2.3 Business Questions

- What is the total sales value generated?
- What is the total profit and overall profit margin?
- How many orders and units were processed?
- Which regions generate the highest sales and profit?
- Which product categories contribute most to sales and profit?
- Which products are the top revenue generators?
- How does sales performance change by month?
- Which customer segments contribute most to sales?
- How can management use these patterns for product, regional and promotional decisions?

CHAPTER 3 — TOOLS & TECHNOLOGIES

3.1 Why Power BI?

Microsoft Power BI was selected because it combines data preparation, relational modelling, DAX calculations and interactive visual analytics in a single environment. It supports slicers, KPI cards, charts, interactive filtering and professional dashboard layouts suitable for management reporting.

Tool / Technology	Purpose
Microsoft Power BI Desktop	Data modelling, DAX, dashboard creation and interactive reporting
Power Query / Power BI Data Preparation	Data type validation and preparation
DAX	Measures, calculated columns and analytical calculations
Excel / Workbook Dataset	Source format for e-commerce transaction data

3.2 Dashboard Design Principles

- Use KPI cards for high-level business measures.
- Use charts that answer a specific business question.
- Avoid unnecessary duplication of visuals.
- Use chronological sorting for dates and months.
- Keep slicers consistent and easy to locate.
- Use descriptive titles so viewers understand each visual.
- Separate descriptive analysis from recommendations.

CHAPTER 4 — DATA PREPARATION & DATA MODEL

4.1 Data Type Validation

The e-commerce dataset was reviewed and prepared before analysis in Power BI. Appropriate data types were assigned to ensure accurate calculations, filtering and visualization.

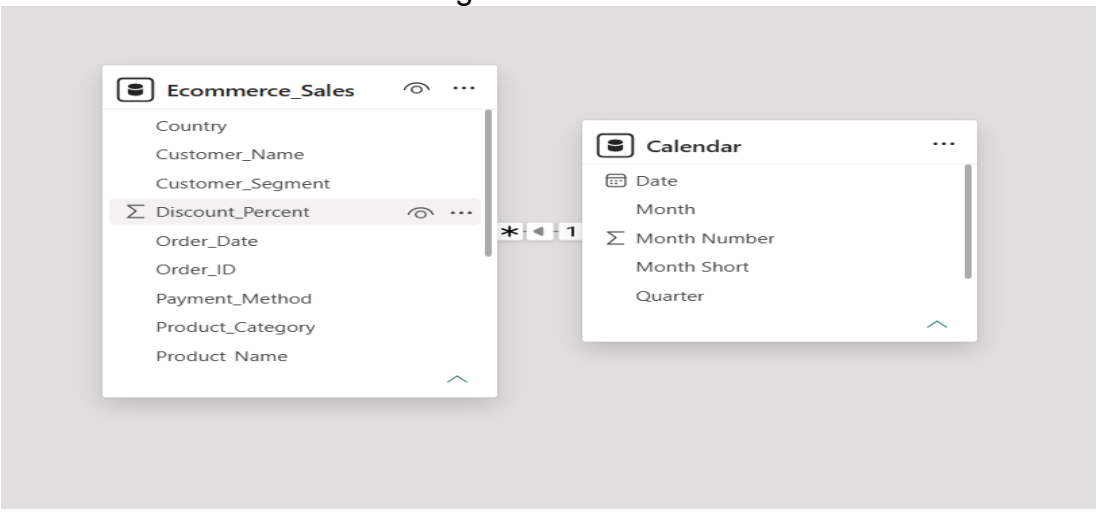
Field	Data Type	Purpose
Order_Date	Date	Time-based analysis
Order_ID	Text / Identifier	Distinct order counting
Customer_Name	Text	Customer analysis
Customer_Segment	Text / Category	Segment analysis
Product_Category	Text / Category	Category analysis
Product_Name	Text / Category	Product analysis
Region	Text / Category	Regional analysis
Country	Text / Category	Geographical analysis
Quantity	Whole Number	Unit volume
Discount_Percent	Decimal Number	Discount analysis
Profit	Decimal Number	Profit calculations
Shipping_Cost	Decimal Number	Shipping analysis
Total_Sales	Decimal Number	Sales calculations
Unit_Price	Decimal Number	Pricing analysis

4.2 Calendar Table

A dedicated Calendar table was created to support consistent time-based analysis. The table contains Date, Year, Month Number, Month, Month Short and Quarter. Month is sorted using Month Number so monthly visuals appear in chronological order.

4.3 Relationship

The data model uses an active many-to-one relationship from Calendar[Date] to Ecommerce_Sales[Order_Date]. This creates a standard date-dimension-to-sales-table structure and enables time-based filtering across the dashboard.



4.4 Calculated / Supporting Columns

Column	Logic / Purpose
Year	YEAR(Calendar[Date])
Month Number	MONTH(Calendar[Date])
Month	FORMAT(Calendar[Date], "MMMM")
Month Short	FORMAT(Calendar[Date], "MMM")
Quarter	"Q" & FORMAT(Calendar[Date], "Q")

4.5 Data Quality Checks

- Checked for duplicate or invalid transaction records.
- Verified numerical fields and appropriate data types.
- Verified Order_Date as a Date field.
- Created and validated the Calendar table.
- Established the active relationship between Calendar and Ecommerce_Sales.
- Verified chronological month sorting.
- Validated DAX measures against source columns.

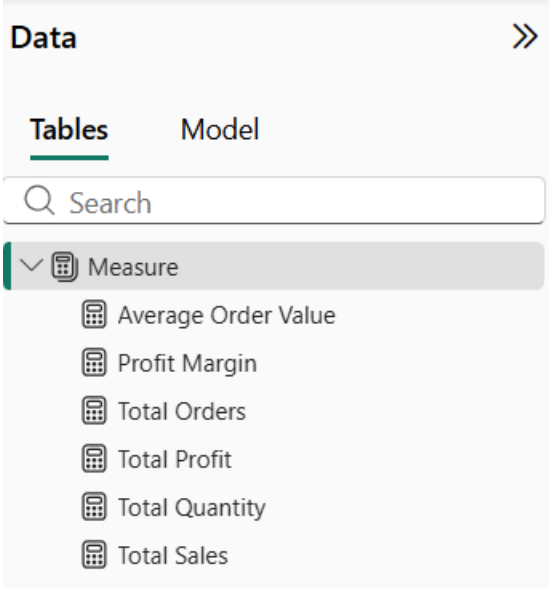
CHAPTER 5 — DAX MEASURES & CALCULATIONS

DAX measures were used instead of hard-coded values so that every KPI responds dynamically to filters and slicers. The following measures form the analytical foundation of the dashboard.

Measure	DAX Formula	Purpose
Total Sales	SUM(Ecommerce_Sales[Total_Sales])	Overall sales value
Total Profit	SUM(Ecommerce_Sales[Profit])	Overall profit
Total Orders	DISTINCTCOUNT(Ecommerce_Sales[Order_ID])	Unique orders
Total Quantity	SUM(Ecommerce_Sales[Quantity])	Units sold
Profit Margin	DIVIDE([Total Profit], [Total Sales], 0)	Profit as % of sales
Average Order Value	DIVIDE([Total Sales], [Total Orders], 0)	Average sales per order

5.1 Core DAX Code

```
Total Sales = SUM(Ecommerce_Sales[Total_Sales])
Total Profit = SUM(Ecommerce_Sales[Profit])
Total Orders = DISTINCTCOUNT(Ecommerce_Sales[Order_ID])
Total Quantity = SUM(Ecommerce_Sales[Quantity])
Profit Margin = DIVIDE([Total Profit], [Total Sales], 0)
Average Order Value = DIVIDE([Total Sales], [Total Orders], 0)
```

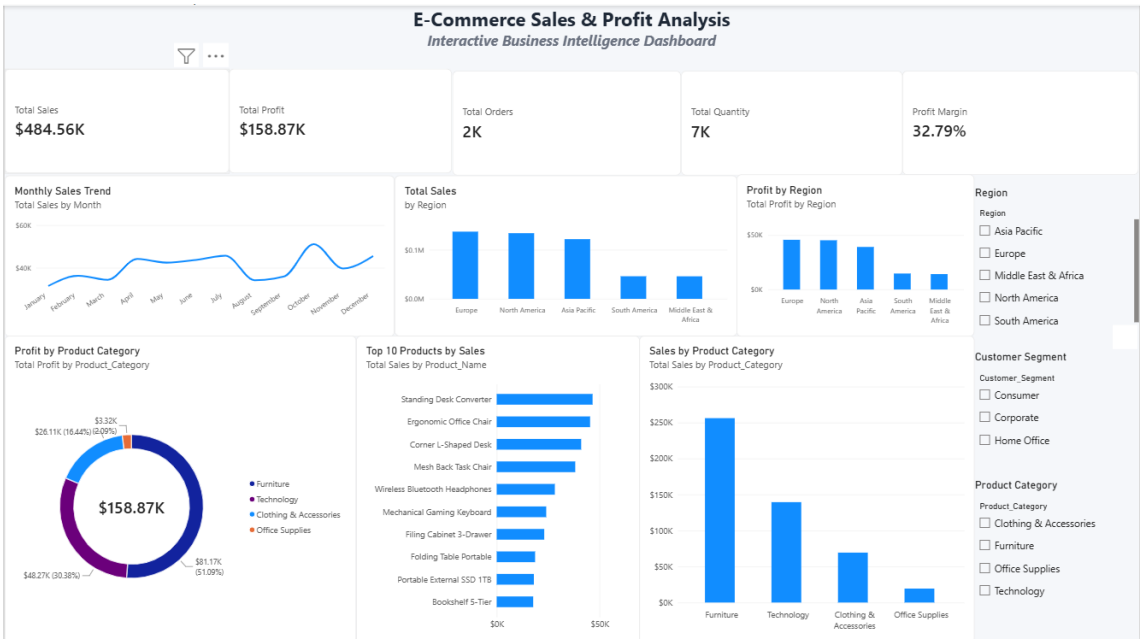


CHAPTER 6 — DASHBOARD DEVELOPMENT

6.1 Executive Overview

The Executive Overview is designed as the management landing page. It presents the most important KPIs first and then provides high-level trends, regional comparisons, product-category analysis and interactive filters.

Visual	Fields / Measure	Purpose
KPI Card	Total Sales	Overall sales value
KPI Card	Total Profit	Overall profit
KPI Card	Total Orders	Order volume
KPI Card	Total Quantity	Units sold
KPI Card	Profit Margin	Profitability ratio
Line Chart	Month + Total Sales	Monthly sales trend
Column Chart	Region + Total Sales	Regional sales comparison
Column Chart	Region + Total Profit	Regional profit comparison
Column Chart	Product Category + Total Sales	Category sales contribution
Bar Chart	Top 10 Product Name + Total Sales	Top revenue products
Donut Chart	Product Category + Total Profit	Profit contribution by category
Slicers	Region, Customer Segment, Product Category	Interactive filtering



6.2 Visual Design Used

The dashboard follows a clean executive-reporting layout. KPI cards are placed at the top, trend and regional visuals are placed in the middle, and product/category analysis is placed in the lower section. Slicers are positioned on the right side to provide interactive filtering.

CHAPTER 7 — ANALYTICAL FINDINGS

The dashboard is designed to produce findings at executive, regional, product and customer-segment levels. Final numerical interpretation should be based on the values visible in the completed Power BI report.

7.1 Executive-Level Findings

- Total Sales provides the overall scale of business revenue.
- Total Profit indicates overall profit generated.
- Total Orders shows transaction volume.
- Total Quantity indicates units sold.
- Profit Margin provides a profitability indicator relative to sales.
- Average Order Value provides a benchmark for typical order value.

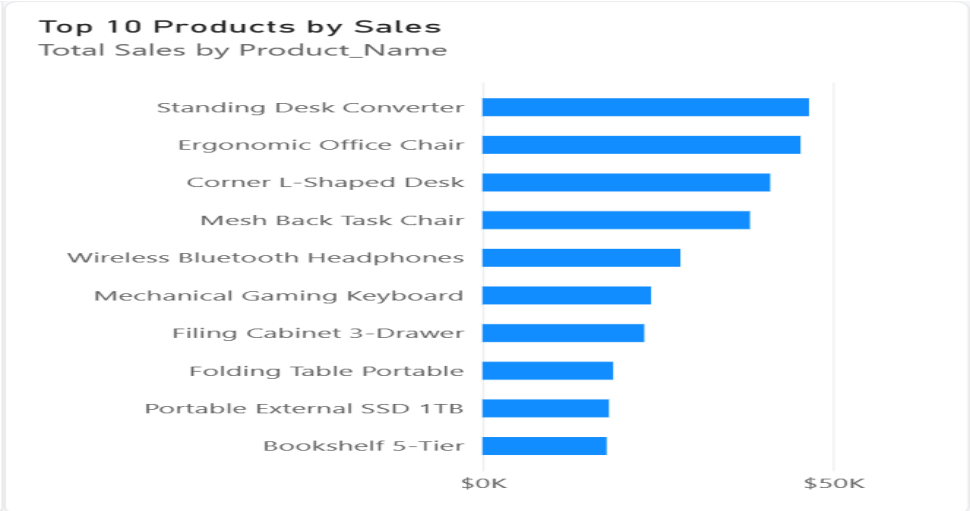
7.2 Regional Findings

The Profit by Region and Total Sales by Region visuals allow management to compare regional contribution. High-performing regions can be studied for successful product mix, customer behaviour and sales strategies, while lower-performing regions can be reviewed for improvement.



7.3 Product and Category Findings

Sales by Product Category and Profit by Product Category identify which categories contribute most strongly to sales and profit. The Top 10 Products by Sales visual highlights individual products that make a significant contribution to overall revenue.



7.4 Time-Based Findings

The Monthly Sales Trend visual helps identify changes in sales across the reporting period. Peaks and declines can be investigated alongside regional, category and customer-segment filters to understand possible patterns.

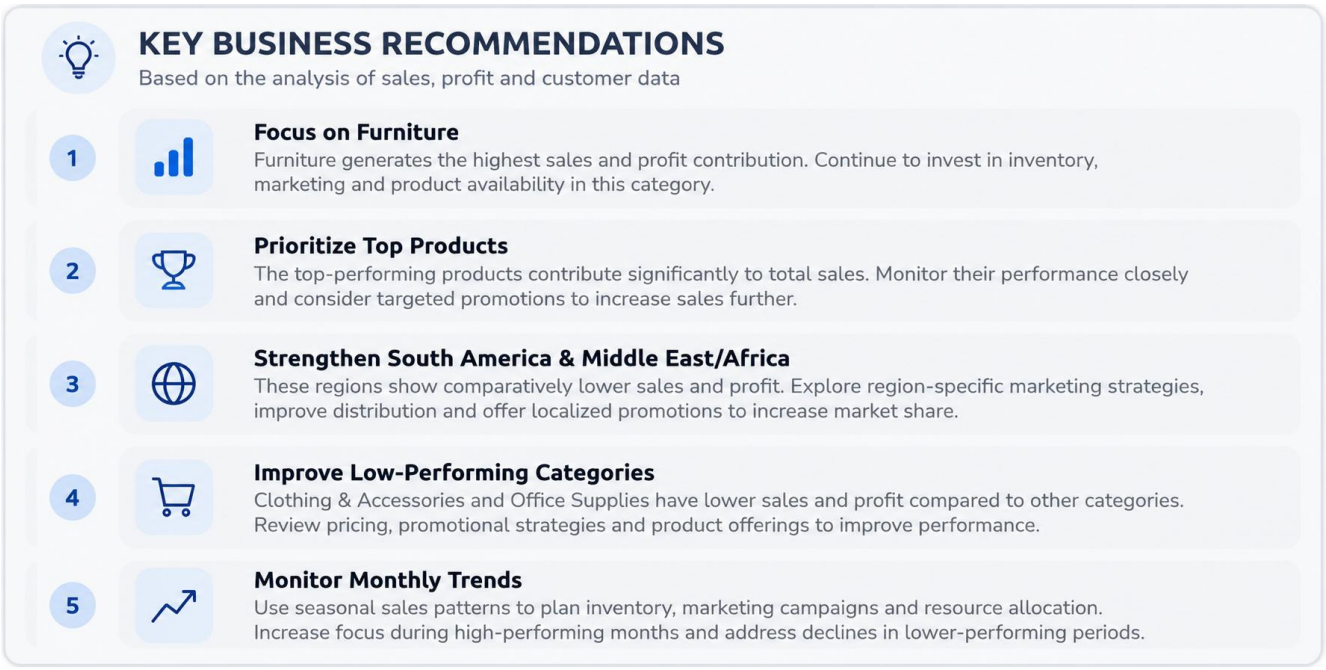


7.5 Important Interpretation Rule

Sales and profit should be considered together when making business decisions. A product or region with high sales is not automatically the most profitable.

CHAPTER 8 — BUSINESS RECOMMENDATIONS

Recommendation	Business Action
Protect high-performing products	Keep high-sales and high-profit products available and visible.
Focus on strong regions	Study high-performing regions and replicate successful practices where appropriate.
Review low-performing regions	Investigate product mix, pricing, customer segments and sales activity.
Promote profitable categories	Use category-level sales and profit results to guide promotional emphasis.
Review low-performing products	Investigate products with low sales or profit before changing or removing them.
Use customer-segment insights	Compare segment performance to improve targeting and promotional planning.
Review discount impact	Compare discounts with sales and profit to ensure promotions create value.
Use profit margin for decisions	Consider profit alongside sales when evaluating performance.



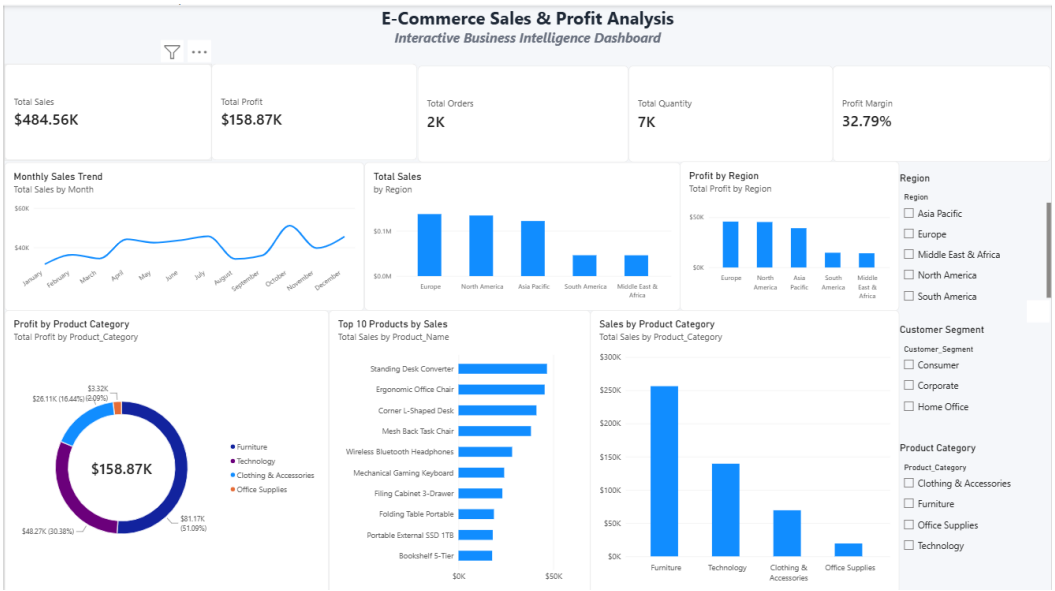
CHAPTER 9 — TESTING & VALIDATION

9.1 Measure Validation

Test	Validation Approach	Status
Total Sales	Compare KPI with SUM(Ecommerce_Sales[Total_Sales]).	Validated
Total Profit	Compare KPI with SUM(Ecommerce_Sales[Profit]).	Validated
Total Orders	Compare with distinct Order_ID count.	Validated
Total Quantity	Compare with SUM(Ecommerce_Sales[Quantity]).	Validated
Profit Margin	Confirm Total Profit / Total Sales.	Validated
Category totals	Check category values against overall totals.	Validated
Month sorting	Confirm months appear chronologically.	Implemented & Validated
Slicers	Confirm visuals respond to selections.	Validated

9.2 Visual Validation

- Confirm all chart titles describe the metric shown.
- Confirm sales charts use Total Sales when aggregation is required.
- Confirm order counts use Total Orders rather than row count.
- Confirm Top 10 Products returns the intended number of products.
- Confirm profit visuals use Total Profit.
- Confirm slicers respond consistently.



CHAPTER 10 — LIMITATIONS & FUTURE SCOPE

10.1 Limitations

- The analysis is limited to fields available in the supplied dataset.
- Customer behavioural analysis is limited to available customer fields.
- The dashboard represents the period contained in the dataset and is not real-time unless a live refresh pipeline is implemented.
- Inventory, returns, competitor pricing and campaign information are not included unless added.
- Profit analysis depends on the supplied Profit field and should not be interpreted as a complete accounting statement.

10.2 Future Enhancements

- Add month-over-month and year-over-year growth measures.
- Add forecasting models for sales and profit.
- Add marketing campaign and promotion data.
- Add customer history for retention and lifetime-value analysis.
- Add inventory data.
- Implement automated refresh.
- Add a decomposition tree for root-cause analysis.

CHAPTER 11 — CONCLUSION

The E-Commerce Sales & Profit Analysis project demonstrates how transactional sales data can be transformed into a structured Business Intelligence solution using Microsoft Power BI. The project establishes a clear analytical model, creates reusable DAX measures and presents interactive dashboards focused on executive performance, regional behaviour, product categories and sales trends.

The Executive Overview provides a concise management view through KPI cards, monthly sales trends, regional comparisons, product-category analysis, Top 10 product ranking and interactive slicers. The model also separates sales performance from profit performance, allowing both measures to be considered when interpreting business results.

Overall, the project provides a practical foundation for e-commerce sales reporting and can be extended through forecasting, customer analytics, inventory information, marketing data and automated refresh capabilities.

REFERENCES

1. Microsoft Power BI Documentation — Power BI Desktop, data modelling, visualizations and report design.

2. Microsoft Learn — DAX documentation and function reference.

3. Source E-Commerce Sales Dataset — https://www.kaggle.com/code/muhammadaammartufail/global-e-commerce-sales-customer-analytics/input?

4. Microsoft Excel / Office Documentation — spreadsheet data handling and workbook formats.

APPENDIX A — DAX CODE

A.1 Core Measures

```
Total Sales = SUM(Ecommerce_Sales[Total_Sales])
Total Profit = SUM(Ecommerce_Sales[Profit])
Total Orders = DISTINCTCOUNT(Ecommerce_Sales[Order_ID])
Total Quantity = SUM(Ecommerce_Sales[Quantity])
Profit Margin = DIVIDE([Total Profit], [Total Sales], 0)
Average Order Value = DIVIDE([Total Sales], [Total Orders], 0)
```

A.2 Calendar Table

```
Calendar =
ADDCOLUMNS(
    CALENDAR(
        MIN(Ecommerce_Sales[Order_Date]),
        MAX(Ecommerce_Sales[Order_Date])
    ),
    "Year", YEAR([Date]),
    "Month Number", MONTH([Date]),
    "Month", FORMAT([Date], "MMMM"),
    "Month Short", FORMAT([Date], "MMM"),
    "Quarter", "Q" & FORMAT([Date], "Q")
)
```